

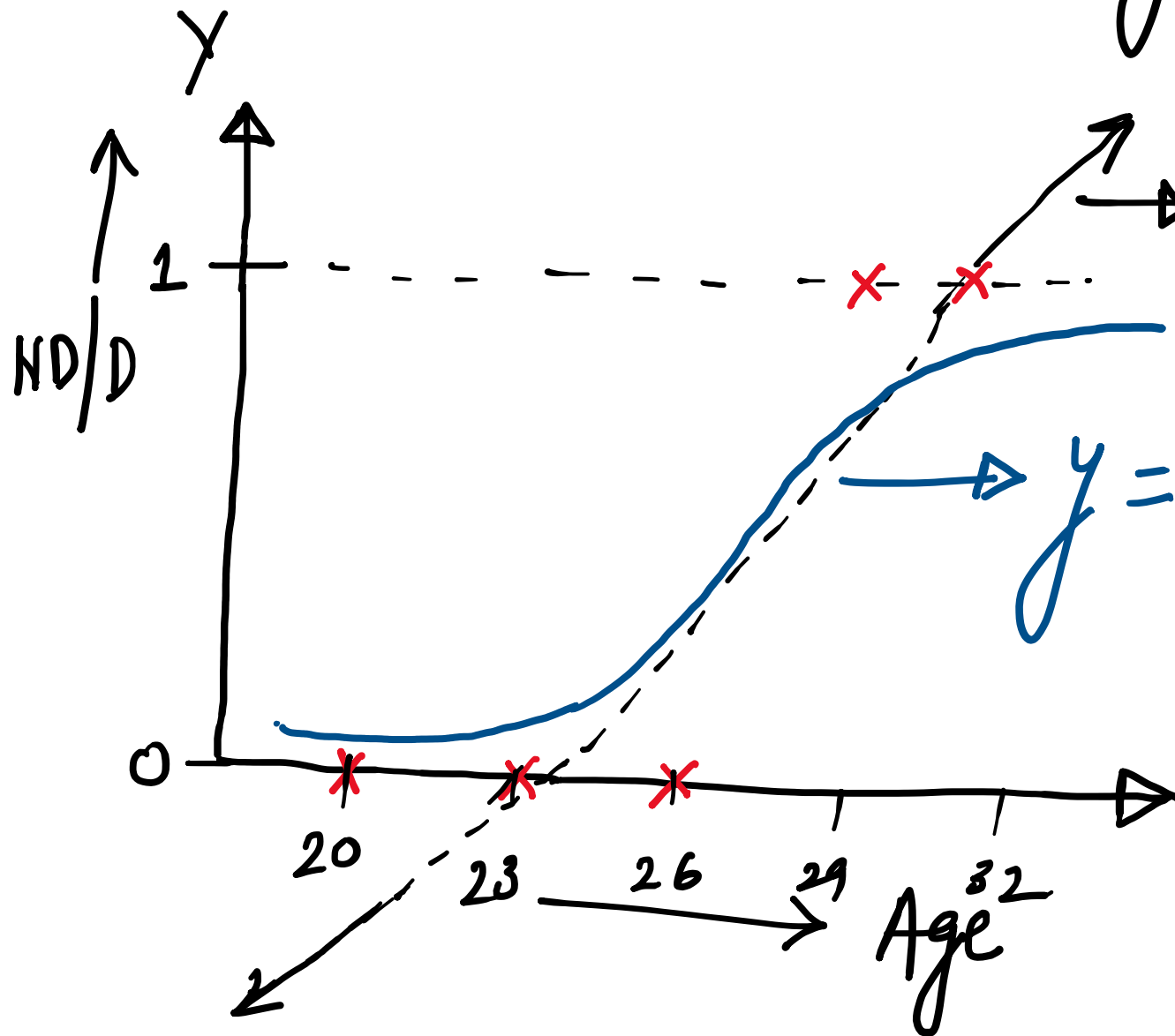
ML



→ Applications :-

- * Approval/Rejection of CC/loan application (fintech)
- * Churn/Non-Churn of Customer (telecom)
- * Customer reviews Sentiment analysis (Positive/Negative/Neutral)
- * Customer will purchase Upgrade course or not (Marketing)

→ Intuition behind logistic regression :- $D \rightarrow 1 (+)$ $ND \rightarrow 0$ (GLM)



$y = mx + c$

$y = f(mx + c)$

Age	D/ND
20	0
22	0
24	0
28	1
30	1

Sigmoid transformⁿ
 or
 logistic transformⁿ (logit)

→ Linear Regression :-

$$y \in [-\infty, \infty]$$

→ { Logistic Regression } :- transformatⁿ (logistic)
or
Sigmoid

$$y \in [0, 1]$$

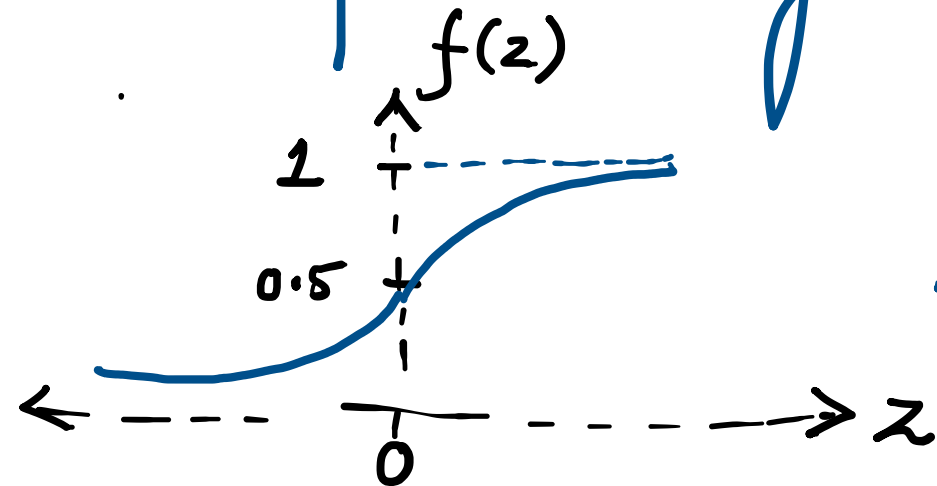
* Generalised linear model

→ Sigmoid f^n :- $f(z)$ | $y = f(mx+c)$

* S-Shaped curve

* $f(z) \in [0, 1]$

* Mathematically,



$$y = \frac{1}{1 + e^{-(mx+c)}}$$

$$f(z) = \frac{1}{1 + e^{-z}}$$

→ final eqⁿ
for logistic
regression

Ques:- Calculate the value of $f(z)$ for $z=1000$!

Sol

$$f(z) = \frac{1}{1+e^{-z}}$$

$$= \frac{1}{1 + \cancel{e^{-1000}} \rightarrow 0} \approx 1$$

Ques:- Calculate the value of $f(z)$ for $z = -1000$!

Sol

$$f(z) = \frac{1}{1 + e^{-z}}$$
$$= \frac{1}{1 + e^{-(-1000)}} = \frac{1}{1 + e^{1000}} = \frac{1}{\text{very big number}}$$
$$\approx 0$$

→ Odds :- Ratio of prob. of any event happening to the prob of same event ^{not} happening.

$$\text{Odds}(X=1) = \frac{P(X=1)}{P(X \neq 1)} = \frac{1/6}{1-1/6} = \underline{\underline{1/5}}$$

$$\text{odds}(X=h) = \frac{P(X=h)}{P(X \neq h)} = \frac{1/2}{1/2} = 1$$

→ log odds :- comparative odds

Logarithm of odds

Mathematics

ML

$$\text{odds} = \frac{p}{1-p}$$

$$\log \text{ odds} = \log \left(\frac{p}{1-p} \right) = \log \left(\frac{y}{1-y} \right)$$

→ Mathematical Derivation :-

$$\log\left(\frac{y}{1-y}\right) = mx + c$$

$$\frac{y}{1-y} = e^{mx+c}$$

} logit f^n
→ log odds follows
linear equation

$$y = (1-y)e^{mx+c}$$

$$y = e^{mx+c} - y e^{mx+c}$$

$$y + y e^{mx+c} = e^{mx+c}$$

$$y(1 + e^{mx+c}) = e^{mx+c}$$

$$y = \frac{e^{mx+c}}{1 + e^{mx+c}}$$

$$y = \frac{\cancel{e^{mx+c}}}{1 + \cancel{e^{mx+c}}} \cdot \frac{e^{mx+c}}{e^{mx+c}}$$

$$y = \frac{1}{\frac{1}{e^{mx+c}} + \frac{e^{mx+c}}{e^{mx+c}}}$$



$$y = \frac{1}{1 + e^{-(mx+c)}}$$

Evaluation Metrics:-

$\begin{cases} 1 \rightarrow +ve \\ 0 \rightarrow -ve \end{cases}$

* Confusion Matrix:- 2×2 matrix

Predictions

	+	-
+	TP	FN
-	FP	TN

$\uparrow$ TP $\rightarrow$ True positives ($+ \rightarrow +$)
 $\uparrow$ TN $\rightarrow$ True negatives ($- \rightarrow -$)
 $\downarrow$ FP $\rightarrow$ False positives ($- \rightarrow +$)
 $\downarrow$ FN $\rightarrow$ False negatives ($+ \rightarrow -$)

Actuals
 P
 $0^- \rightarrow 0^- - 0^- - 0^- \}$ TN
 $0^- \rightarrow 1^+ - 0^- - 1^+ \}$ FP
 $1^+ \rightarrow 0^- - 1^+ - 0^- \}$ FN
 $1^+ \rightarrow 1^+ - 1^+ - 1^+ \}$ TP

→ Accuracy :-

$$A = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

$$A = \frac{TP + TN}{TP + FP + TN + FN}$$

→ Recall / Sensitivity / True Positive Rate (TPR) :-

$$R = \frac{\text{Correctly predicted +ve}}{\text{Total actual +ve}}$$

$$R = \frac{TP}{TP + FN}$$

→ Precision :-

$$P = \frac{\text{Correctly predicted +ve}}{\text{Total } \underline{\underline{\text{predicted +ve}}}}$$

$$P = \frac{TP}{TP + FP}$$

→ F1-Score :- (Combination of Precision & Recall)

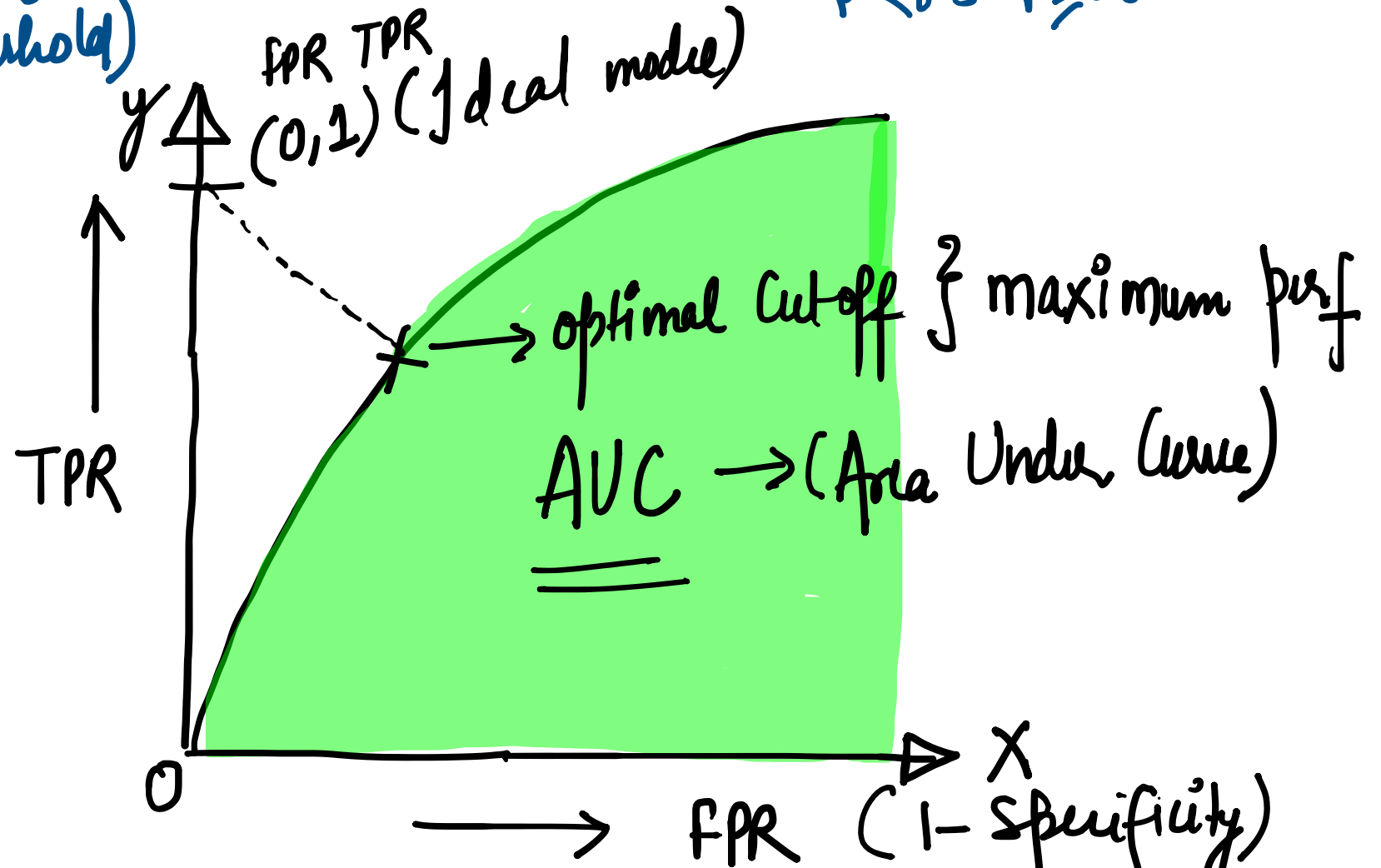
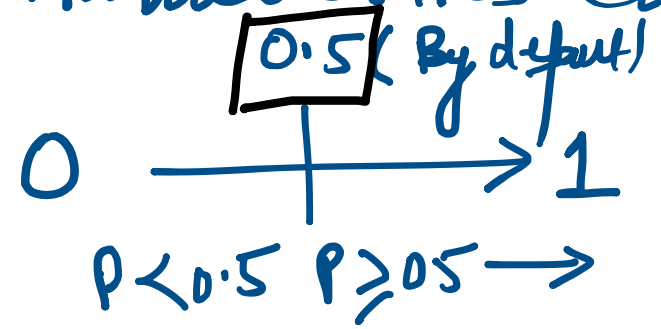
$$\text{F1-Score} = \frac{2 \times P \times R}{P + R}$$

* Harmonic mean of precision & recall.

→ ROC-AUC :- Receiver Operating Characteristics Curve

* Performance of classification model (AUC)

* model cutoff (threshold)



$$\begin{array}{l} 1 \quad p_1 \\ 2 \quad p_2 \\ 3 \quad p_3 \end{array} \left. \vphantom{\begin{array}{l} 1 \\ 2 \\ 3 \end{array}} \right\} \max(p_1, p_2, p_3) \longrightarrow \text{final class}$$